

Tasca S5.01

Consultes amb MongoDB

Treballarem amb una base de dades que conté col·leccions relacionades amb una aplicació d'entreteniment cinematogràfic.

Duràs a terme algunes consultes que et demana el client/a, el qual està mesurant si seràs capaç o no de fer-te càrrec de la part analítica del projecte vinculat amb la seva base de dades.

Nivell 1

Tasca inicial:

Crea una base de dades amb **MongoDB** utilitzant com a col·leccions els arxius adjunts:

- **users:** Emmagatzema informació d'usuaris/es, incloent-hi noms, emails i contrasenyes xifrades.
- **theatres:** Conté dades de cinemes, com ID, ubicació (direcció i coordenades geogràfiques).
- **sessions:** Guarda sessions d'usuari, incloent-hi ID d'usuari i tokens JWT per a l'autenticació.
- **movies:** Inclou detalls de pel·lícules, com a trama, gèneres, durada, elenc, comentaris, any de llançament, directors, classificació i premis.
- **comments:** Emmagatzema comentaris d'usuaris/es sobre pel·lícules, amb informació de l'autor/a del comentari, ID de la pel·lícula, text del comentari i la data.

Base de dados: cinema

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel shows a connection to 'localhost:27017' with a database named 'Cinema'. The 'Cinema' database is expanded, showing collections: 'comments', 'movies', 'sessions', 'theaters', and 'users'. The main panel displays a table of these collections with their properties.

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
comments	-	401.41 kB	14.29 MB	50K	284.00 B	1	45.06 kB
movies	-	4.10 kB	37.62 MB	24K	1.60 kB	1	4.10 kB
sessions	-	4.10 kB	540.00 B	1	540.00 B	1	4.10 kB
theaters	-	4.10 kB	349.83 kB	1.6K	223.00 B	1	4.10 kB
users	-	4.10 kB	29.57 kB	185	159.00 B	1	4.10 kB

Exercici 1

Tasques:

- A. Mostra els 2 primers comentaris que hi ha en la base de dades.
- B. Quants usuaris tenim registrats?
- C. Quants cinemes hi ha en l'estat de Califòrnia?
- D. Quin va ser el primer usuari/ària en registrar-se?
- E. Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

Nivell 1 - Exercici 1

- A. Els 2 primers comentaris:
ordenem la col·lecció comments per data i mostrem els dos primers registres.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' sidebar lists the connection 'localhost:27017' with a tree view containing 'Cinema', 'comments', 'movies', 'sessions', 'theaters', 'users', 'admin', 'config', and 'local'. The 'comments' collection is selected. The main panel displays a query editor with the following settings:

- Project:** { field: 0 }
- Sort:** {"date":1} (highlighted in yellow)
- Collation:** { locale: 'simple' }
- Index Hint:** { field: -1 } or "indexName"
- Max Time MS:** 60000
- Skip:** 0
- Limit:** 2 (highlighted in yellow)

Buttons at the top right include 'Explain', 'Reset', 'Find' (highlighted in green), and 'Options'. Below the query editor are buttons for 'ADD DATA', 'BULK', 'EXPORT DATA', and 'EXPORT CODE'. The results section shows two documents:

```
_id: ObjectId('5a9427648b0beeb695e67e')
name: "Mercedes Tyler"
email: "mercedes_tyler@fakegmail.com"
movie_id: ObjectId('573a1399f29313caabcec3ce')
text: "Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cu_"
date: 1970-01-01T01:07:09.000+00:00
```

```
_id: ObjectId('5a9427658b0beeb6969efe')
name: "Jon Snow"
email: "kit_harington@gameofthron.es"
movie_id: ObjectId('573a13a4f29313caabd117df')
text: "Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur_"
date: 1970-01-01T09:37:49.000+00:00
```

B. Número d'usuaris registrats: 185

Fem una agregació. Al pas 1 escollim \$count per comptar tots els registres de la col·lecció users.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel shows a connection to 'localhost:27017' with a database named 'Cinema'. The 'users' collection is selected. The main panel shows the 'Aggregations' tab for the 'users' collection, which contains 185 documents. The aggregation pipeline is defined as follows:

```
1 // **
2 * Provide the field name for the count.
3 */
4 { '$count': 'Num_usuaris:' }
```

The output preview shows the result of the aggregation:

```
{ 'Num_usuaris': 185 }
```

Nivell 1 - Exercici 1

C. Número de cinemes a l'estat de Califòrnia: 169

Opció 1: fem filtre: {"location.address.state": "CA"} i mirem número de registres.

Opció 2: fem agregació amb un \$match i un \$count.

The screenshot displays the MongoDB Compass interface with the following components:

- Left Panel (Connections):** Shows a tree view of the database structure. The 'theaters' collection is selected under the 'Cinema' database on the 'localhost:27017' connection.
- Top Panel (Query Builder):** Contains a filter query: `{ "location.address.state": "CA" }`. It also shows document counts: 25 / 1 - 25 of 169.
- Center Panel (Document Viewer):** Displays two sample documents from the 'theaters' collection, each with fields like `_id`, `theaterId`, and `location`.
- Right Panel (Aggregations):** Shows an aggregation pipeline with two stages:
 - Stage 1 (\$match):** The query is `{ "location.address.state": "CA" }`. The output preview shows a sample of 10 documents.
 - Stage 2 (\$count):** The query is `'Número de cinemes'`. The output preview shows the result: `Número de cinemes : 169`.

Nivell 1 - Exercici 1

- D. Primer usuari registrat:
ordenem la col·lecció users per _id i després mostrem un registre.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' sidebar lists a connection to 'localhost:27017' with a database named 'Cinema'. The 'Cinema' database contains several collections: 'comments', 'movies', 'sessions', 'theaters', 'users', 'admin', 'config', and 'users'. The 'users' collection is selected.

The main query editor displays the following query:

```
{ field: 'value' }
```

Buttons for 'Explain', 'Reset', 'Find', and 'Options' are visible. The 'Sort' field is set to `{ "-id": 1 }`. The 'Collation' is `{ locale: 'simple' }`. The 'Index Hint' is `{ field: -1 } or "indexName"`. The 'Max Time MS' is set to 60000. The 'Skip' is 0 and the 'Limit' is 1.

Below the query editor, there are buttons for 'ADD DATA', 'BULK', 'EXPORT DATA', 'EXPORT CODE', and a help icon.

The results section shows one document:

```
{
  "_id": ObjectId('59b99db4cfa9a34dcd7885b6'),
  "name": "Ned Stark",
  "email": "sean_bean@gameofthron.es",
  "password": "$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74cr1J1Vu"
}
```


E. Número de pel·lícules de comèdia: 7024

Primer fem filtre a la col·lecció movies: {genres: 'Comedy'}. Després mirem número de registres.

CONNECTIONS (1)

Search connections

localhost:27017

Cinema

comments

movies

sessions

theaters

users

admin

config

local

{genres: 'Comedy'}

Generate query

Explain

Res

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

25

1 - 25 of 7024

```

_id: ObjectId('573a1390f29313caabcd4803')
plot: "Cartoon figures announce, via comic strip balloons, that they will mov..."
genres: Array (3)
  0: "Animation"
  1: "Short"
  2: "Comedy"
runtime: 7
cast: Array (1)
num_mflix_comments: 1
poster: "https://m.media-amazon.com/images/M/MV5BYzg2NjNhNTctMjUxMi00ZWU4LWI3Zj..."
title: "Winsor McCay, the Famous Cartoonist of the N.Y. Herald and His Moving ..."
fullplot: "Cartoonist Winsor McCay agrees to create a large set of drawings that ..."
languages: Array (1)
released: 1911-04-08T00:00:00.000+00:00
directors: Array (2)
writers: Array (2)
awards: Object
lastupdated: "2015-08-29 01:09:03.030000000"
year: 1911
imdb: Object
countries: Array (1)
type: "movie"
tomatoes: Object

```

Exercici 2

Tasca:

- Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

Script:

- `{ $and: [{ year: 1932, $or: [{ genres: "Drama" }, { languages: "French" }] } ] }`

Nivell 1 - Exercici 2

🕒

{ \$and: [{ year: 1932, \$or: [{ genres: "Drama" }, { languages: "French" }] }] }

[Generate query](#) 

Explain

 ADD DATA

 BULK

 EXPORT DATA

 EXPORT CODE



25

1 - 18 of 18



```
_id: ObjectId('573a1391f29313caabcd9458')
plot: "A young artist draws a face at a canvas on his easel. Suddenly the mou..."
runtime: 55
rated: "UNRATED"
▶ cast: Array (4)
num_mflix_comments: 1
poster: "https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWEtYjlmYi00NjA4LTk4ZW..."
title: "The Blood of a Poet"
lastupdated: "2015-09-16 13:13:05.537000000"
▼ languages: Array (1)
  0: "French"
released: 2010-05-20T00:00:00.000+00:00
▶ directors: Array (1)
▶ writers: Array (1)
▶ awards: Object
  year: 1932
▶ imdb: Object
▶ countries: Array (1)
  type: "movie"
▶ tomatoes: Object
```

```
_id: ObjectId('573a1392f29313caabcd99a3')
plot: "Junta is hated by the people in the village where she lives, especiall..."
▼ genres: Array (3)
  0: "Drama"
  1: "Fantasy"
  2: "Mystery"
runtime: 85
▶ cast: Array (4)
poster: "https://m.media-amazon.com/images/M/MV5BNTQ1NTMzMtQ0ODIyYS00MTAxLWE1NT..."
```

Exercici 3

Tasca:

- Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

Script:

- `{ $and: [{ countries: "USA", "awards.wins": { $gte: 5, $lte: 9 }, year: { $gte: 2012, $lte: 2014 } } ] }`

Nivell 1 - Exercici 3

 `{&and:[{countries: "USA", "awards.wins":{$gte:5,$lte:9}, year:{$gte:2012,$lte:2014}` [Generate query](#)  [Explain](#)

[+ ADD DATA](#) [BULK](#) [EXPORT DATA](#) [EXPORT CODE](#) 

25 1 - 25 of 166 

```
_id: ObjectId('573a13acf29313caabd29366')
fullplot: "The manager of the negative assets sector of Life magazine, Walter Mit..."
imdb: Object
  year: 2013
  plot: "When his job along with that of his co-worker are threatened, Walter t..."
  genres: Array (3)
  rated: "PG"
  metacritic: 54
  title: "The Secret Life of Walter Mitty"
  lastupdated: "2015-08-31 00:10:51.747000000"
  languages: Array (3)
  writers: Array (3)
  type: "movie"
  tomatoes: Object
    poster: "https://m.media-amazon.com/images/M/MV5BODYwNDYxNDk1Nl5BMl5BanBnXkFtZT..."
    num_mflix_comments: 1
    released: 2013-12-25T00:00:00.000+00:00
  awards: Object
    wins: 6
    nominations: 13
    text: "6 wins & 13 nominations."
  countries: Array (2)
    0: "USA"
    1: "Canada"
  cast: Array (4)
  directors: Array (1)
  runtime: 114
```

Nivell 2

Exercici 1:

- Compta quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

Exercici 2:

- Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

Nivell 2 - Exercici 1

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

Cinema

comments

movies

sessions

theaters

users

admin

config

local

localhost:27017 > Cinema > comments

Documents 50K Aggregations Schema Indexes 1 Validation

Smatch \$count

Generate aggregation

Untitled - modified

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

Stage 1 \$match

```
1 /**
2  * query: The query in MQL.
3  */
4 {
5   "email": { "$regex": "@gameofthron\\.es$", "$options": "i" }
6 }
```

Output preview after \$match stage (Sample of 10 documents)

```
_id: ObjectId('5a9427648b0beebe69579d0')
name: "Talisa Maegyr"
email: "oona_chaplin@gameofthron.es"
movie_id: ObjectId('573a1390f29313caabcd41b...')
text: "Rem itaque ad sit rem voluptatibus.
      Ad fugiat maxime illum optio iure ..."
date: 1998-08-22T11:45:03.000+00:00
```

Stage 2 \$count

```
1 /**
2  * Provide the field name for the count.
3  */
4 'Número usuarios:'
```

Output preview after \$count stage (Sample of 1 document)

```
Número usuarios: : 22841
```

Nivell 2 - Exercici 2

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

Cinema

comments

movies

sessions

theaters

users

admin

config

local

localhost:27017 > Cinema > theaters

Documents 1.6K Aggregations Schema Indexes 1 Validation

Match \$group

Generate aggregation + ? Explain

Untitled - modified SAVE + CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES TEXT

Stage 1 \$match

```
1 /**
2  * query: The query in MQL.
3  */
4 {
5   "location.address.state": "DC"
6 }
```

Output preview after \$match stage (Sample of 3 documents)

_id: ObjectId('59a47286cfa9a3a73e51e785')

theaterId: 1092

location: Object

Stage 2 \$group

```
1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5 {
6   _id: "$location.address.zipcode",
7   total_cinemes: {$sum: 1}
8 }
```

Output preview after \$group stage

_id: "20016"

total_cinemes: 1

STAGE OUTPUT

Sample of 3 documents

_id: "20016"

total_cinemes: 1

_id: "20010"

total_cinemes: 1

_id: "20002"

total_cinemes: 1

Nivell 3

Exercici 1:

- Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

Exercici 2:

- Mostra en un mapa la ubicació de tots els teatres de la base de dades.

Nivell 3 - Exercici 1

Script: `{\$and:[{directors: "John Landis", "imdb.rating":{\$gte:7.5,\$lte:8} }]}`

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' sidebar shows a connection to 'localhost:27017' with a database named 'Cinema'. The 'movies' collection is selected. The main area displays a query: `{\$and:[{directors: "John Landis", "imdb.rating":{\$gte:7.5,\$lte:8} }]}`. Below the query bar are buttons for 'ADD DATA', 'BULK', 'EXPORT DATA', and 'EXPORT CODE'. The results pane shows two documents. The first document is for the movie 'Animal House', directed by John Landis, with an IMDb rating of 7.6. The second document is for the movie 'Jake Blues', also directed by John Landis.

Query: `{\$and:[{directors: "John Landis", "imdb.rating":{\$gte:7.5,\$lte:8} }]}`

Results:

```
{
  "_id": ObjectId('573a1397f29313caabce6d94'),
  "fullplot": "Faber College has one frat house so disreputable it will take anyone. ...",
  "imdb": {
    "rating": 7.6,
    "votes": 84834,
    "id": 77975
  },
  "year": 1978,
  "plot": "At a 1962 college, Dean Vernon Wormer is determined to expel the entire...",
  "genres": [
    "R"
  ],
  "rated": "R",
  "metacritic": 82,
  "title": "Animal House",
  "lastupdated": "2015-09-13 00:02:47.803000000",
  "languages": [
    "English"
  ],
  "writers": [
    "Meyerowitz, David"
  ],
  "tomatoes": {
    "poster": "https://m.media-amazon.com/images/M/MV5BM2M2ZDA4MTYtOGRjMi00OTg5LWI1ZT...",
    "num_mflix_comments": 1,
    "released": "1978-07-28T00:00:00.000+00:00"
  },
  "awards": {
    "countries": [
    ],
    "cast": [
    ],
    "directors": [
      "John Landis"
    ],
    "runtime": 109
  }
}
```

```
{
  "_id": ObjectId('573a1397f29313caabce76f7'),
  "plot": "Jake Blues, just out from prison, puts together his old band to save t...",
  "genres": [
    "Comedy"
  ]
}
```

Nivell 3 - Exercici 2

